

1(a)(i)	to allow, oxygen / air, to enter for respiration or because, bacterium / <i>B. subtilis</i> , respire aerobically / AW ;	1
1(a)(ii)	as, population growth of bacteria / bacterial growth, increases / AW, the, cloudier / more turbid, the culture broth or measuring the, turbidity / cloudiness, by eye against a cross in background / with a colorimeter / turbidity meter / colour chart ;	1
1(b)	replace antibiotic with, (distilled) water / nutrient solution ;	1
1(c)	<i>can improve:</i> add, catalase / peroxidase / manganese(IV) oxide ; <i>explain why:</i> results / effect / killing, <i>B. subtilis</i> / bacteria , is only due to, honey / MGO or results / effect / killing, <i>B. subtilis</i> / bacteria, is not due to hydrogen peroxide ;	2
1(d)	<i>description or table or diagram for 10%:</i> 1 Correct method to dilute stock (honey) solution to 10% ; <i>table</i> 2 At least five dilutions including 10% all with correct volumes of stock and nutrient solution including units ;	2
1(e)	<i>independent variable</i> MGO / Manuka / honey, concentration ; <i>dependent variable</i> turbidity ;	2

Question	Answer	Marks
1(f)	<i>predict:</i> lower concentration, tubes / AW, will be, turbid / cloudy or higher concentration, tubes / AW, will be, clear / not cloudy or the lower the concentration the more, turbid / cloudy, the tubes / AW ; ora <i>explain:</i> MGO / (Manuka) honey, inhibits the growth of / kills, <i>B. subtilis</i> / <i>bacteria</i> or lower concentrations have less, ability / MGO, to, inhibit the growth of / kill, <i>B. subtilis</i> / <i>bacteria</i> ; ora	2

Question	Answer	Marks												
1(g)	any five from: 1 same / stated, volume of, culture / broth used ; 2 same / stated, volume of Manuka honey ; 3 mix / stir, honey (concentrations) and, culture / broth (prior to starting) ; 4 method to measure turbidity ; 5 <i>idea that:</i> write down / note / record, which concentration(s) the broth, is clear / is clearest ; any two from 6–8 ;; 6 method of maintaining temperature (10°C – 50°C if stated) 7 same / stated, time (left in incubator) (12–48 hours if stated) 8 use a buffer to maintain pH (pH 4–10 if stated) 9 repeat at least twice / 3 replicates and finding mean ; 10 named hazard and risk and precaution ; <table border="1"> <thead> <tr> <th>Hazard</th><th>Risk</th><th>Precaution</th></tr> </thead> <tbody> <tr> <td>Live cultures / bacterium / <i>B. subtilis</i></td><td>Allergy / Infection / irritation / toxic</td><td>Wear gloves / goggles / mask / PPE</td></tr> <tr> <td>Honey</td><td>Allergy</td><td>Wear gloves / goggles / mask / PPE</td></tr> <tr> <td>Broth</td><td>Irritant / allergy</td><td>Wear gloves / goggles / mask / PPE</td></tr> </tbody> </table>	Hazard	Risk	Precaution	Live cultures / bacterium / <i>B. subtilis</i>	Allergy / Infection / irritation / toxic	Wear gloves / goggles / mask / PPE	Honey	Allergy	Wear gloves / goggles / mask / PPE	Broth	Irritant / allergy	Wear gloves / goggles / mask / PPE	5
Hazard	Risk	Precaution												
Live cultures / bacterium / <i>B. subtilis</i>	Allergy / Infection / irritation / toxic	Wear gloves / goggles / mask / PPE												
Honey	Allergy	Wear gloves / goggles / mask / PPE												
Broth	Irritant / allergy	Wear gloves / goggles / mask / PPE												
1(h)(i)	there is no difference in the effect / effectiveness, of Manuka (honey) and Germania (honey) ;	1												

Question	Answer	Marks
1(h)(ii)	yes yes ; no	1
1(h)(iii)	<i>results:</i> Manuka honey, kills more bacteria / is more effective (than Germania) ; ora statistical analysis: first three / 1, 2 and 3, are significantly different or last two / 4 and 5, not significantly different against MRSA or 1 is only significant at $p = 0.05$ or 2 / 3, are significant at the, $p = 0.01$ / $p = 0.001$;	2